

PSCPR Grading Rubric

Problem-Solving & Critical Path Reasoning — The Three Types of Expert

Rubric

	Type 1 Expert <i>Known Knowns & Known Unknowns</i> Observation & Analysis	Type 2 Expert <i>Unknown Knowns</i> Inference	Type 3 Expert <i>Unknown Unknowns</i> Exploration
Essence	<i>Competent within established frameworks. Does what should be obvious and identifies what needs to be found.</i>	<i>Recognizes implicit knowledge others overlook. Makes inferential connections the problem doesn't spell out.</i>	<i>Explores genuinely novel territory. Discovers what remains unknown and questions what the answer implies.</i>
Look for	- Identifies what is given - States what is being asked - Selects appropriate equation(s) - Executes correctly with units	- Recognizes unstated constraints - Connects concepts across topics - States and justifies assumptions - Reduces degrees of freedom	- Checks dimensions and units - Tests limiting cases - Verifies order of magnitude - Notes implications or anomalies
Full credit	Correct setup and result. Work is visible and organized.	Identifies what the problem doesn't say. Connects or simplifies beyond the obvious.	Independently verifies the result. Asks "does this make sense?" and answers it.
Partial	Right approach, arithmetic or algebra error. Missing units but method sound.	Sees something deeper but doesn't fully develop it. Assumption stated but not justified.	Attempts a check but incomplete or inconclusive.
No credit	No work shown, wrong method without justification, or answer only.	No reasoning beyond what was given. Followed a recipe without understanding why.	No verification attempted. Answer presented as final without reflection.

Applying to Question Types

Not every question can elicit all three types. The **ceiling** is the highest type a question can reasonably reach. Grade to the ceiling; reward students who exceed it. ○ = possible but not expected.

Question Format	T1	T2	T3	Default %	How to Grade
Multiple choice / true-false / fill-in / signature	✓	—	—	100	Binary or near-binary.
Single-step calculation	✓	—	—	100	Method + result. Partial for right setup, wrong number.
Multi-step problem	✓	✓	—	60 / 40	Credit inferential connections beyond plug-and-chug.
Short answer / explain reasoning	✓	✓	○	60 / 40	Restating a formula = T1. Explaining why = T2.
Multi-part derivation or proof	✓	✓	✓	40 / 30 / 30	Full rubric. Reward sanity checks and limiting cases.

Default % shows how to split a question's points across types (T1 / T2 / T3). Adjust per assignment as needed.

Edge Cases

Right answer, no work shown: Minimal credit. On Type 1—only questions, the answer *is* the work. On anything higher, work is required.

Wrong answer, strong reasoning: Substantial credit. A wrong answer to a proper question is informative. Award full T2 and T3 credit where earned; deduct within T1.

Student exceeds the ceiling: Award bonus credit. This is the behavior we want to cultivate.

Cascading errors: Don't penalize the same mistake twice. Deduct once, then evaluate remaining work on its own logic.

"Show your work": Means show your *thinking*—a diagram, stated assumptions, a justification, a sanity check. Not every algebraic step.

Type 1 = they followed the procedure. Type 2 = they understood the structure. Type 3 = they can think independently.

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Extended Dimensions — Projects, Lab Reports & Papers

The core rubric (page 1) evaluates reasoning on bounded problems. Projects, lab reports, and papers require additional dimensions. The same Type 1 → 2 → 3 framework applies to each.

What to Look For

Dimension	Type 1 Expert <i>Known Knowns & Known Unknowns</i>	Type 2 Expert <i>Unknown Knowns</i>	Type 3 Expert <i>Unknown Unknowns</i>
Reasoning Technical analysis and problem-solving	• Correct application of theory • Calculations executed properly • Appropriate methods selected	• Assumptions identified and justified • Connections made across concepts • Model limitations recognized	• Results verified independently • Limiting cases explored • Open questions identified
Communication Clarity, structure, presentation of ideas	• Follows required structure • Figures and tables labeled • Writing is clear and correct • Results clearly stated	• Narrative connects sections logically • Figures chosen to reveal insight, not just display data • Readable without the author present	• Distinguishes observation from inference from speculation • Uncertainty communicated honestly • Depth appropriate to audience
Methodology Design, data handling, reproducibility	• Procedure followed correctly • Data recorded completely • Equipment and setup documented • Units and sig figs consistent	• Sources of error identified • Controls and variables distinguished • Data processing choices justified	• Systematic vs random error discussed • Alternative approaches considered • Reproducibility assessed
Professional Practice Integrity, standards, collaboration	• Submitted on time and complete • Proper citations and references • Contributions documented (if team) • Safety protocols followed	• Sources evaluated, not just cited • Peer feedback incorporated • Ethical considerations noted	• Reflects on what worked and didn't • Proposes concrete next steps • Intellectually honest about limitations

A lab report that describes what happened is Type 1. One that explains why is Type 2. One that asks what it means is Type 3.

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Assignment Profiles — How to Weight the Dimensions

Dimension Weights

How much each dimension counts depends on what the assignment asks the student to do. These are defaults — adjust per assignment.

	Reasoning	Communication	Methodology	Prof. Practice
Project	40%	20%	25%	15%
Lab Report	30%	20%	40%	10%
Paper / Report	30%	35%	20%	15%
Presentation / Poster	25%	40%	15%	20%

Projects weight Reasoning highest because the student defines the problem and makes design choices. Professional Practice matters more here than in lab reports because projects involve scope management, collaboration, and deliverable quality.

Lab Reports weight Methodology highest because the experiment is often prescribed — the question is whether the student can execute, document, and interpret rigorously.

Papers and Reports weight Communication alongside Reasoning because the primary deliverable is a written argument.

Presentations and Posters weight Communication highest. The work may already be done — the question is whether the student can convey it clearly and engage with questions.

What Each Type Looks Like

Concrete examples of Type 1, 2, and 3 performance for each assignment category.

Assignment Type	Type 1 Emphasis	Type 2 Emphasis	Type 3 Emphasis
Project	Implementation works. Requirements met. Deliverables complete.	Design choices justified. Trade-offs articulated. Scope managed thoughtfully.	Identifies what they'd do differently. Proposes extensions. Honest about limitations.
Lab Report	Procedure followed. Data recorded. Calculations correct. Format complete.	Error sources identified. Data processing justified. Results connected to theory.	Systematic vs random error distinguished. Results questioned. Anomalies addressed.
Paper / Report	Sources cited. Structure followed. Claims supported by evidence.	Argument developed logically. Sources synthesized, not just listed. Counterarguments addressed.	Original analysis or insight. Identifies gaps in existing work. Distinguishes established from speculative.
Presentation / Poster	Content accurate. Visuals readable. Time managed. Required elements present.	Story arc connects the work. Visuals chosen for insight. Anticipates audience questions.	Engages with implications. Handles Q&A thoughtfully. Acknowledges uncertainty openly.

The core rubric evaluates how a student thinks. These pages evaluate how they apply, communicate, and practice that thinking.

Print and attach to student work

Assignment: _____ Total Points: _____

Coursework

Question	Ceiling	T1 %	T2 %	T3 %	Score	Notes
e.g. Prob 1	T2	90%	75%	—	8.3/10	Missed friction
/	/	/	/	/	/	/
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Dimension	Weight	T1 %	T2 %	T3 %	Score	Notes
Reasoning	/	/	/	/	/	/
Communication	/	/	/	/	/	/
Methodology	/	/	/	/	/	/
Prof. Practice	/	/	/	/	/	/
		/	/	/	/	

FINAL SCORE: _____ / _____